Working Notes for $D_{s1} \to D_s\gamma$ in Light-Cone QCD Sum Rules

Ds gamma project

July 1, 2026

Abstract

These notes are a living derivation for the light-cone QCD sum-rule calculation of $D_{s1}(2460) \to D_s\gamma$ and $D_{s1}(2536) \to D_s\gamma$. The purpose is pedagogical as well as practical: each step is written so that a master's student can follow the choices, conventions, and algebra before the final paper-level formulas are assembled.

1 Conventions

We use the mostly-minus metric

$$g_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1), \tag{1}$$

and

$$\{\gamma_\mu, \gamma_\nu\} = 2g_{\mu\nu}, \qquad \sigma_{\mu\nu} = \frac{\mathrm{i}}{2}[\gamma_\mu, \gamma_\nu], \qquad \gamma_5 = \mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3. \tag{2}$$

The Levi-Civita convention is

$$\epsilon^{0123} = +1, \qquad \epsilon_{0123} = -1, \tag{3}$$

so that

$$\mathrm{Tr}[\gamma_5\gamma_\mu\gamma_\nu\gamma_\alpha\gamma_\beta] = -4\mathrm{i}\,\epsilon_{\mu\nu\alpha\beta}. \tag{4}$$

Our Fourier convention for a propagator is

$$S(x) = \int \frac{d^4k}{(2\pi)^4}\,\mathrm{e}^{-\mathrm{i}k\cdot x}\,\widetilde{S}(k). \tag{5}$$

2 Currents and Momentum Assignment

The external momenta are

$$P = p + q, \tag{6}$$

where P is the initial axial-meson momentum, p is the final pseudoscalar D_s momentum, and q is the real-photon momentum.

The pseudoscalar interpolating current is

$$J_{D_s}(x) = \bar{s}(x)\,\mathrm{i}\gamma_5\,Q(x). \tag{7}$$

For the axial channel we use two independent currents,

$$J^{(0)}_{A\mu}(x) = \bar{s}(x)\gamma_\mu\gamma_5 Q(x), \tag{8}$$

$$J^{(0)}_{B\mu}(x) = \frac{\mathrm{i}}{m_Q + m_s}\,\bar{s}(x)\sigma_{\mu\alpha}P^\alpha\gamma_5 Q(x). \tag{9}$$

The tensor current contracts with P^α, not p^α, because it interpolates the initial axial state. The factor $1/(m_Q + m_s)$ restores the same mass dimension as the axial-vector current.

3 Correlation Function

For a generic axial current $\Gamma_\mu \in \{\gamma_\mu\gamma_5,\, \mathrm{i}\sigma_{\mu\alpha}P^\alpha\gamma_5/(m_Q+m_s)\}$ we study

$$\Pi_\mu(p,q) = \mathrm{i}\int d^4x\, \mathrm{e}^{\mathrm{i}p\cdot x}\langle\gamma(q,\varepsilon)|T\{\bar{s}(x)\Gamma_\mu Q(x)\,\bar{Q}(0)\mathrm{i}\gamma_5 s(0)\}|0\rangle. \tag{10}$$

The target Lorentz structure for the $1^+ \to 0^-\gamma$ transition is

$$(\eta\cdot\varepsilon^*)(p\cdot q) - (\eta\cdot q)(p\cdot\varepsilon^*). \tag{11}$$

On the correlator side we therefore project onto an equivalent transverse structure built from the open axial index and the photon polarization.

4 Wick Contraction and OPE Bookkeeping

The OPE is separated into a hard-photon sector and a soft-photon sector:

$$\Pi_\mu^{\mathrm{OPE}} = \Pi_\mu^{\mathrm{hard}} + \Pi_\mu^{\mathrm{soft}}. \tag{12}$$

This split is important because the two terms describe different physics and must not be double counted.

4.1 Hard photon emission

In the hard sector the photon is emitted perturbatively from either the heavy quark or the strange quark. After Wick contraction one obtains

$$\Pi_\mu^{\mathrm{hard}} = \mathrm{i}N_c\int d^4x\, \mathrm{e}^{\mathrm{i}p\cdot x}\mathrm{Tr}_D\Big[\Gamma_\mu S_Q^\gamma(x)\,\mathrm{i}\gamma_5\, S_s^0(-x) + \Gamma_\mu S_Q^0(x)\,\mathrm{i}\gamma_5\, S_s^\gamma(-x)\Big], \tag{13}$$

where $N_c = 3$ and Tr_D denotes the Dirac trace. The free massive propagator is

$$S_q^0(x) = \int \frac{d^4k}{(2\pi)^4}\,\mathrm{e}^{-\mathrm{i}k\cdot x}\frac{\mathrm{i}(\not{k}+m_q)}{k^2-m_q^2}. \tag{14}$$

We keep explicit powers of m_s, including possible m_s^2 terms.

The first-order electromagnetic background-field correction for a generic quark flavor q is

$$S_q^\gamma(x) = -\mathrm{i}e_q\int\frac{d^4k}{(2\pi)^4}\,\mathrm{e}^{-\mathrm{i}k\cdot x}\int_0^1 du\left[\frac{\not{k}+m_q}{2(m_q^2-k^2)^2}F^{\alpha\beta}(ux)\sigma_{\alpha\beta} + \frac{ux_\alpha}{m_q^2-k^2}F^{\alpha\beta}(ux)\gamma_\beta\right]. \tag{15}$$

In Eq. (13) this generic formula is used twice:

$$S_Q^\gamma(x) = S_q^\gamma(x)\big|_{q\to Q,\, m_q\to m_Q,\, e_q\to e_Q}, \tag{16}$$

$$S_s^\gamma(x) = S_q^\gamma(x)\big|_{q\to s,\, m_q\to m_s,\, e_q\to e_s}. \tag{17}$$

Thus the hard sector contains both heavy-line and strange-line electromagnetic background-field insertions. For a real photon we write

$$F^{\alpha\beta}(ux) = \mathrm{i}(\varepsilon^\alpha q^\beta - \varepsilon^\beta q^\alpha)\mathrm{e}^{\mathrm{i}uq\cdot x}, \tag{18}$$

with the overall phase kept consistent with the Fourier convention.

Equation (15) is not a vacuum-condensate correction. It is the quark propagator in an external electromagnetic field and is included in the base analysis. What we do not include in the base hard sector are the ordinary SVZ-style condensate-expanded propagator terms such as $\langle\bar{s}s\rangle$, $\langle\bar{s}g_s\sigma Gs\rangle$, or $\langle G^2\rangle$ propagator corrections.

4.2 Equivalent momentum-space vertex form

For FeynCalc trace checks it is often simpler to use the equivalent explicit photon-vertex representation. With photon index ν,

$$N_{Q,\mu\nu} = \mathrm{Tr}_D[\Gamma_\mu(\not{k} + \not{q} + m_Q)\gamma_\nu(\not{k} + m_Q)\mathrm{i}\gamma_5(\not{k} - \not{p} + m_s)]\,, \tag{19}$$

$$N_{s,\mu\nu} = \mathrm{Tr}_D[\Gamma_\mu(\not{k} + m_Q)\mathrm{i}\gamma_5(\not{k} - \not{p} + m_s)\gamma_\nu(\not{k} - \not{p} + \not{q} + m_s)]\,. \tag{20}$$

These numerators are accompanied by the denominators

$$D_Q = [(k+q)^2 - m_Q^2][k^2 - m_Q^2][(k-p)^2 - m_s^2], \tag{21}$$

$$D_s = [k^2 - m_Q^2][(k-p)^2 - m_s^2][(k-p+q)^2 - m_s^2]. \tag{22}$$

The explicit-vertex form and the background-field form represent the same hard emission physics. We must not add both as independent contributions.

4.3 Soft photon emission: two-particle photon DAs

In the soft sector the photon is represented by its light-cone distribution amplitudes. Contracting only the heavy quark gives

$$\Pi_\mu^{\mathrm{soft},2p} = \mathrm{i}\int d^4x\, \mathrm{e}^{\mathrm{i}p\cdot x}[\Gamma_\mu S_Q^0(x)\mathrm{i}\gamma_5]_{\alpha\beta}\langle\gamma(q,\varepsilon)|\bar{s}_\alpha(x)s_\beta(0)|0\rangle. \tag{23}$$

Fierz rearrangement converts the open spinor indices to a basis of Dirac bilinears,

$$\bar{s}_\alpha(x)s_\beta(0) = -\frac{1}{4}\sum_i(\Gamma_i)_{\beta\alpha}\bar{s}(x)\Gamma_i s(0), \tag{24}$$

so that

$$\Pi_\mu^{\mathrm{soft},2p} = -\frac{\mathrm{i}}{4}\int d^4x\, \mathrm{e}^{\mathrm{i}p\cdot x}\sum_i \mathrm{Tr}_D[\Gamma_\mu S_Q^0(x)\mathrm{i}\gamma_5\Gamma_i]\,\langle\gamma(q,\varepsilon)|\bar{s}(x)\Gamma_i s(0)|0\rangle. \tag{25}$$

The photon matrix elements are then expressed in terms of two-particle photon DAs. Parameters such as $\chi\langle\bar{s}s\rangle$ enter here as part of the photon wavefunction normalization; they are not condensate corrections to the hard propagator.

4.4 Soft photon emission: three-particle photon DAs

To include three-particle photon DAs we must also allow one background gluon to come from the heavy-quark propagator,

$$S_Q^G(x) = -\mathrm{i}g_s\int\frac{d^4k}{(2\pi)^4}\,\mathrm{e}^{-\mathrm{i}k\cdot x}\int_0^1 du\left[\frac{\not{k} + m_Q}{2(m_Q^2 - k^2)^2}G^{\alpha\beta}(ux)\sigma_{\alpha\beta} + \frac{ux_\alpha}{m_Q^2 - k^2}G^{\alpha\beta}(ux)\gamma_\beta\right]. \tag{26}$$

This term is not a gluon-condensate correction. It is an operator insertion that combines with photon matrix elements of the form

$$\langle\gamma(q,\varepsilon)|\bar{s}(x)\Gamma_i g_s G_{\alpha\beta}(ux)s(0)|0\rangle, \tag{27}$$

which define the three-particle photon DAs.

5 Staged Calculation Plan

We proceed in two stages.

Stage 1. Include the hard photon emission terms and the two-particle photon DAs:

$$\Pi_\mu^{\text{Stage 1}} = \Pi_\mu^{\text{hard}} + \Pi_\mu^{\text{soft},2p}. \tag{28}$$

This stage keeps all explicit m_s terms and omits ordinary condensate-expanded propagators.

Stage 2. Add the three-particle photon DA contributions generated by S_Q^G and the corresponding light-cone matrix elements:

$$\Pi_\mu^{\text{Stage 2}} = \Pi_\mu^{\text{Stage 1}} + \Pi_\mu^{\text{soft},3p}. \tag{29}$$

This staging keeps the derivation auditable: first we verify the core hard and two-particle DA structures, then we add the more involved background-gluon terms.

6 Current FeynCalc Artifacts

The script `scripts/step2_current_building_blocks.wl` checks the basic Dirac conventions and defines the current vertices. The script `scripts/step2_hard_photon_numerators.wl` computes numerator-level hard-photon traces in the explicit photon-vertex representation. These traces are stored in `outputs/step2_hard_photon_numerators.txt`.